Oracle SQL – Employees Exercises

17 Queries | Oracle HR Sample Schema

Important: These solutions use Oracle SQL and the standard HR `EMPLOYEES` table. The column name is `EMPLOYEE_ID` (not EMP_ID).

1. Find employees whose employee ID is between 100 and 105

```
SELECT *
FROM employees
WHERE employee_id BETWEEN 100 AND 105;
```

Explanation: BETWEEN includes both boundary values: 100 and 105.

2. Find employees whose IDs are in the specified list

```
SELECT *
FROM employees
WHERE employee_id IN (151, 152, 153, 154, 155);
```

Explanation: IN checks whether the value matches any item in the list.

3. Find employees whose first name starts with P or p

```
SELECT *
FROM employees
WHERE UPPER(first_name) LIKE 'P%';
```

Explanation: UPPER makes the comparison case-insensitive; % represents any sequence of characters.

4. Find employees whose first name ends with A or a

```
SELECT *
FROM employees
WHERE UPPER(first_name) LIKE '%A';
```

Explanation: The % before A allows any characters before the final letter A.

5. Find employees whose first name contains A or a

```
SELECT *
FROM employees
WHERE UPPER(first_name) LIKE '%A%';
```

Explanation: The % symbols on both sides mean A can appear anywhere in the name.

6. Find employees whose third character in first name is E or e

```
SELECT *
FROM employees
WHERE UPPER(SUBSTR(first_name, 3, 1)) = 'E';
```

Explanation: SUBSTR(first_name, 3, 1) returns one character starting at position 3.

7. Find employees who do not have a manager

```
SELECT *
FROM employees
WHERE manager_id IS NULL;
```

Explanation: Use IS NULL to test for NULL; = NULL does not work in Oracle.

8. Find employees who have a manager

```
SELECT *
FROM employees
WHERE manager_id IS NOT NULL;
```

Explanation: IS NOT NULL returns rows where manager_id contains a value.

9. Insert an employee without a manager

```
INSERT INTO employees (
employee_id, first_name, last_name, email,
hire_date, job_id, manager_id
)
VALUES (
999, 'Ahmed', 'Yasser', 'AHMED999',
SYSDATE, 'IT_PROG', NULL
);
```

Explanation: Expected message: 1 row created. Assumes employee_id 999 and email AHMED999 are unused and IT_PROG exists. Additional mandatory columns or constraints may require changes.

10. Find employees whose job ID is AD_VP or IT_PROG

```
SELECT *
FROM employees
WHERE job_id IN ('AD_VP', 'IT_PROG');
```

Explanation: Character values must be enclosed in single quotation marks.

11. Sort employees by last name in ascending order

```
SELECT *
FROM employees
ORDER BY last_name ASC;
```

Explanation: ASC is the default sort order, but is included for clarity.

12. Sort employees by hire date in descending order

```
SELECT *
FROM employees
ORDER BY hire_date DESC;
```

Explanation: DESC places the most recent hire dates first.

13. Sort by department ID ascending, then salary descending

```
SELECT *
FROM employees
ORDER BY department_id ASC, salary DESC;
```

Explanation: Rows are grouped by department order; higher salaries appear first within each department.

14. Display last names in lowercase

```
SELECT LOWER(last_name) AS last_name
FROM employees;
```

Explanation: LOWER converts text to lowercase.

15. Display first names in uppercase

```
SELECT UPPER(first_name) AS first_name
FROM employees;
```

Explanation: UPPER converts text to uppercase.

16. Display first and last names in title case

```
SELECT INITCAP(first_name) AS first_name,
INITCAP(last_name) AS last_name
FROM employees;
```

Explanation: INITCAP capitalizes the first letter of each word and lowercases the remaining letters.

17. Find employees whose last name is Smith, regardless of case

```
SELECT *
FROM employees
WHERE UPPER(last_name) = 'SMITH';
```

Explanation: Matches Smith, SMITH, smith, and other case variations.

Useful Oracle SQL Notes

- BETWEEN includes the start and end values.
- LIKE 'P%' means starts with P; LIKE '%A' means ends with A; LIKE '%A%' means contains A.
- IS NULL and IS NOT NULL test NULL values.
- LOWER, UPPER, and INITCAP change text capitalization.
- SELECT results depend on the data in your EMPLOYEES table.

Official reference: Oracle SQL Language Reference